

Microprogrammed control unit

- ❑ Control Unit
- ❑ Use of Data Path
- ❑ Conditions in Control Signal Logic
- ❑ Generation of the logic for PCin and Gra
- ❑ Control Unit Details
- ❑ Microprogrammed control unit

- Is an alternative to a hardwired control unit
- Logic of the control unit is specified by a microprogram
 - A microprogram consists of a sequence of instructions in a microprogramming language
 - These are very simple instructions that specify micro-operations
- A microprogrammed control unit is a relatively simple logic circuit that is capable of
 - Sequencing through microinstructions and
 - Generating control signals to execute each microinstruction

Basic Definitions of Microprogrammed Control Unit

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- ☐ Control Unit Details
- ☐ Microporgrammed control unit

- Control signals
 - That are to be generated at a given time step (i.e., to be simultaneously active)
- Micro-operations
 - are the functional, or atomic, operations of a processor
 - The events of any instruction cycle can be described as a sequence of such micro-operations
- Microinstruction
 - a set of micro-operations occurring at one time
 - These micro operations are stored in as Control signals in control word
- Microprogram
 - Is a sequence of micro instructions
 - The collection of control words that implement an macroinstruction
 - Also called as “Firmware”

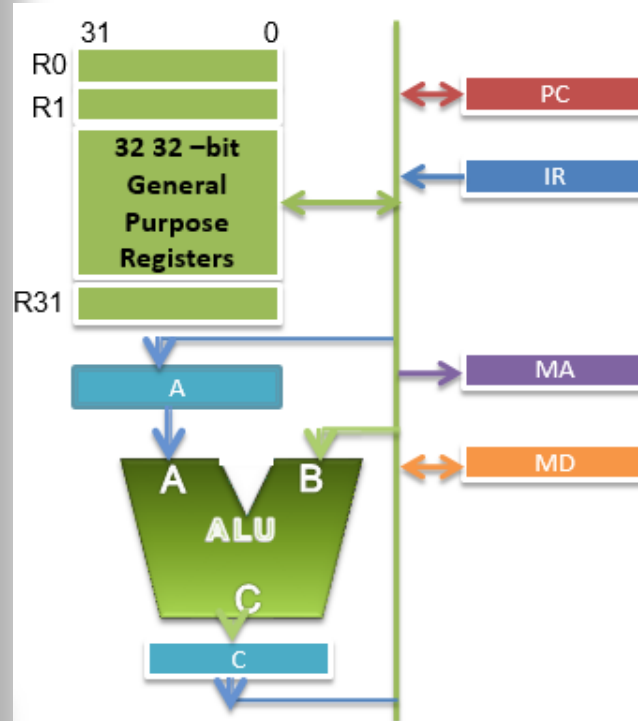
Micro Operations

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- ❑ Microporgrammed control unit
- ❑ Basic definition of MCU
- ❑ Micro operations

Abstract RTN:

(IR $\leftarrow$ M[PC]: PC $\leftarrow$ PC + 4; instruction_execution);
instruction_execution := (• • •
add (:= op= 12) $\rightarrow$ R[ra] $\leftarrow$ R[rb] + R[rc]:

Concrete RTN:

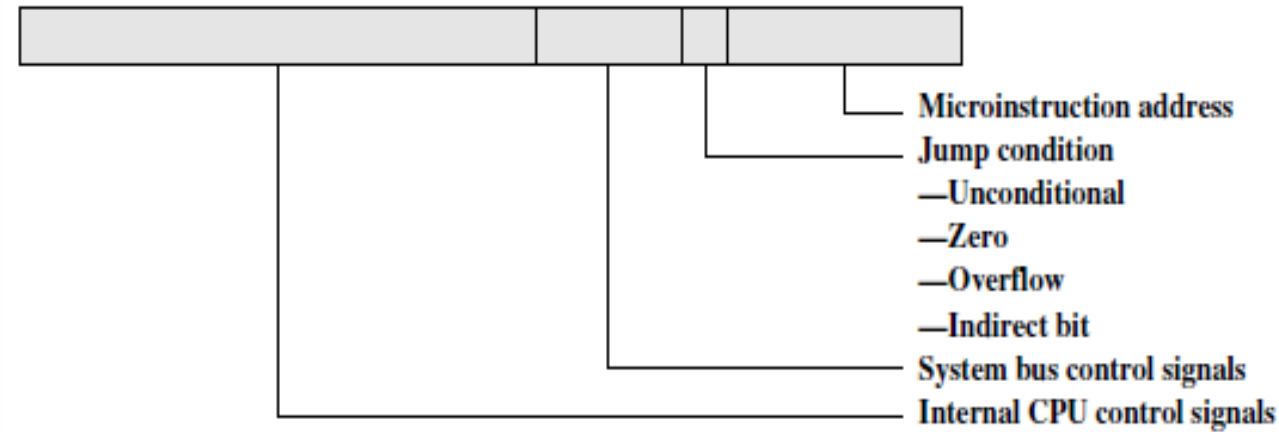


Step	Concrete RTN/ Micro Operations
T0.	MA $\leftarrow$ PC: C $\leftarrow$ PC + 4;
T1.	MD $\leftarrow$ M[MA]; PC $\leftarrow$ C;
T2.	IR $\leftarrow$ MD;
T3.	A $\leftarrow$ R[rb];
T4.	C $\leftarrow$ A + R[rc];
T5.	R[ra] $\leftarrow$ C;

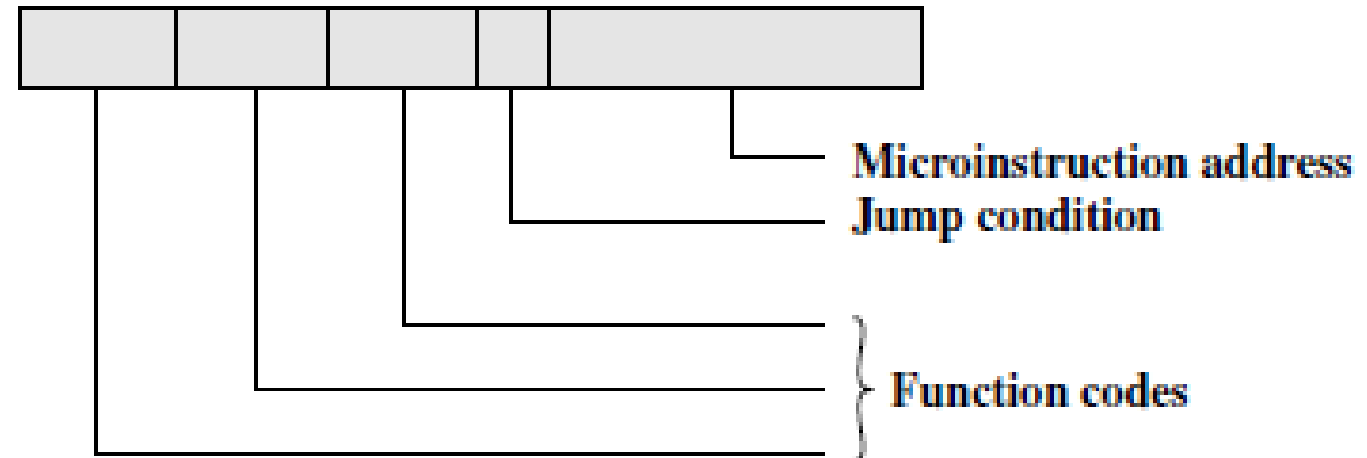
Typical Microinstruction Formats

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- ☐ Micro operations
- ☐ Typical microinstruction function

Horizontal microinstruction



Vertical microinstruction



Steps in interpretation of microinstruction

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- ☐ Typical microinstruction function
- ☐ Steps in interpretation of microinstruction

- To execute this microinstruction,
 - turn on all the control lines indicated by a 1 bit;
 - leave off all control lines indicated by a 0 bit.
 - The resulting control signals will cause one or more micro-operations to be performed
- If the condition indicated by the condition bits is false, execute the next microinstruction in sequence.
- If the condition indicated by the condition bits is true, the next microinstruction to be executed is indicated in the address field.

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Conditions in Control
Signal Logic
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Microporgrammed
control unit

- Control memory
- Used to store control words or micro instructions
- The microinstructions in each routine are to be executed sequentially

Microprogramming

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Microporgrammed
control unit

- The term microprogram was first coined by M.V.Wilkes in the early 1950s
- Wilkes proposed an approach to control unit design that was organized and systematic and avoided the complexities of a hardwired implementation
- Microprogramming became a popular technique for implementing the control unit of CISC processors

Microprogramming

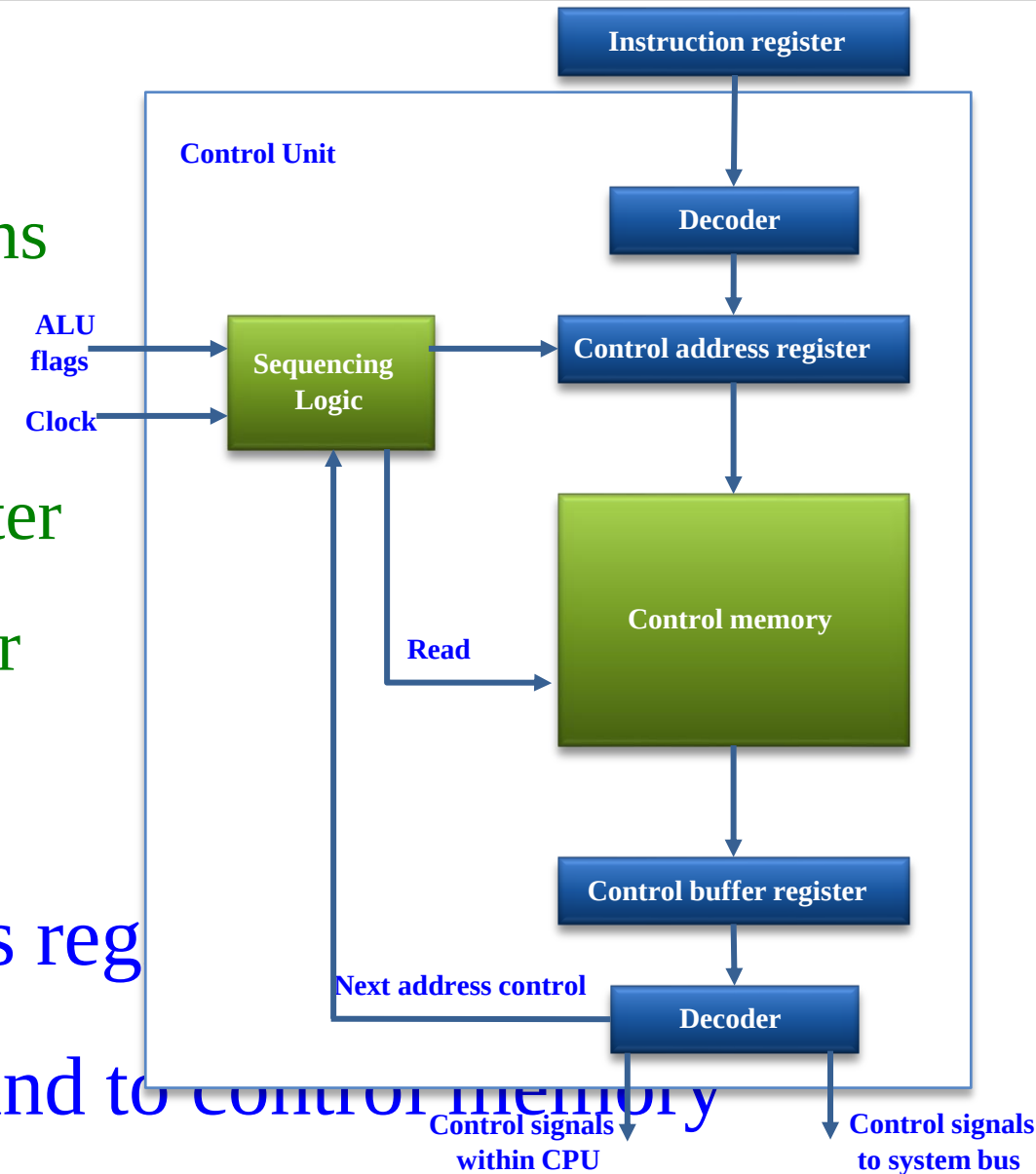
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- Microprogrammed control unit

- Micro program instruction fetch & execution resembles the fetching and execution of a normal program.
- Micro programmed control units are more costly to manufacture than hardware units due to presence of control memory and its access circuitry.
- In 1951, “Maurice Wilkes” published the first paper on this technique, which he called microprogramming.

Key Elements of Control Unit

- ❑ Control Unit
- ❑ Use of Data Path Conditions in Control Signal Logic
- ❑ Generation of the logic for PCin and Gra
- ❑ Control Unit Details Microporgrammed control unit
- ❑ Key Elements of control unit

- Control memory
 - Store microinstructions
- Registers
 - Control address register
 - Control buffer register
- Sequencing logic
- Loads control address reg
- Issues READ command to control memory



Key Elements of control unit

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- Loads control address register
- Issues READ command to control memory
- Functioning of Microprogrammed Control Unit

Steps in function of micro-programmed control unit

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- ☐ Key Elements of control unit
- ☐ Steps in function of Microporgrammed control unit

1. To execute an instruction, the sequencing logic unit issues a READ command to the control memory.
2. The word whose address is specified in the control address register is read into the control buffer register.
3. The content of the control buffer register generates control signals and next address information for the sequencing logic unit.
4. The sequencing logic unit loads a new address into the control address register based on the next-address information from the control buffer register and the ALU flags.

All this happens during one clock pulse

Microcode Execution

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- Microporgrammed control unit
- ☐ Microcode execution

- Op-code is decoded.
- Microinstructions are retrieved from control memory (control address register and the decoder serve as the address register and selection mechanism or control unit).
- The control address register locates the microinstruction to be retrieved from control memory.

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- The microinstruction register holds the retrieved microinstruction - micro opcode and address of the next microinstruction in the control memory
- Current microinstruction is executed.
- The address of the next microinstruction is entered into the control memory to retrieve the next microinstruction.
- If all microinstructions were executed, then store next op-code of conventional instruction in the control address register, if not, execute remaining microinstruction.

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- Conditional jumps are implemented by letting the states of some conditional flip-flops modify the address of the next microinstruction to be retrieved.

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- ❑ Control store/ memory

Step

Control Sequence

T0.

$PC_{out}, MA_{in}, Inc4, C_{in}, Read$

T1.

$C_{out}, PC_{in}, Wait$

T2.

MD_{out}, IR_{in}

add

Step

Control Sequence

T3.

Grb, R_{out}, A_{in}

T4.

$Grc, R_{out}, ADD, C_{in}$

T5.

$C_{out}, Gra, R_{in}, End$

st

Step

Control Sequence

T3.

Grb, BA_{out}, A_{in}

T4.

$c2_{out}, ADD, C_{in}$

T5.

C_{out}, MA_{in}

T6.

$Gra, R_{out}, MD_{in}, Write$

T7.

$Wait, End$

addi

Control Sequence

Grb, R_{out}, A_{in}

$c2_{out}, ADD, C_{in}$

$C_{out}, Gra, R_{in}, End$

shr

Control Sequence

$c1_{out}, Ld$

$n = 0 \rightarrow (Grc, R_{out}, Ld)$

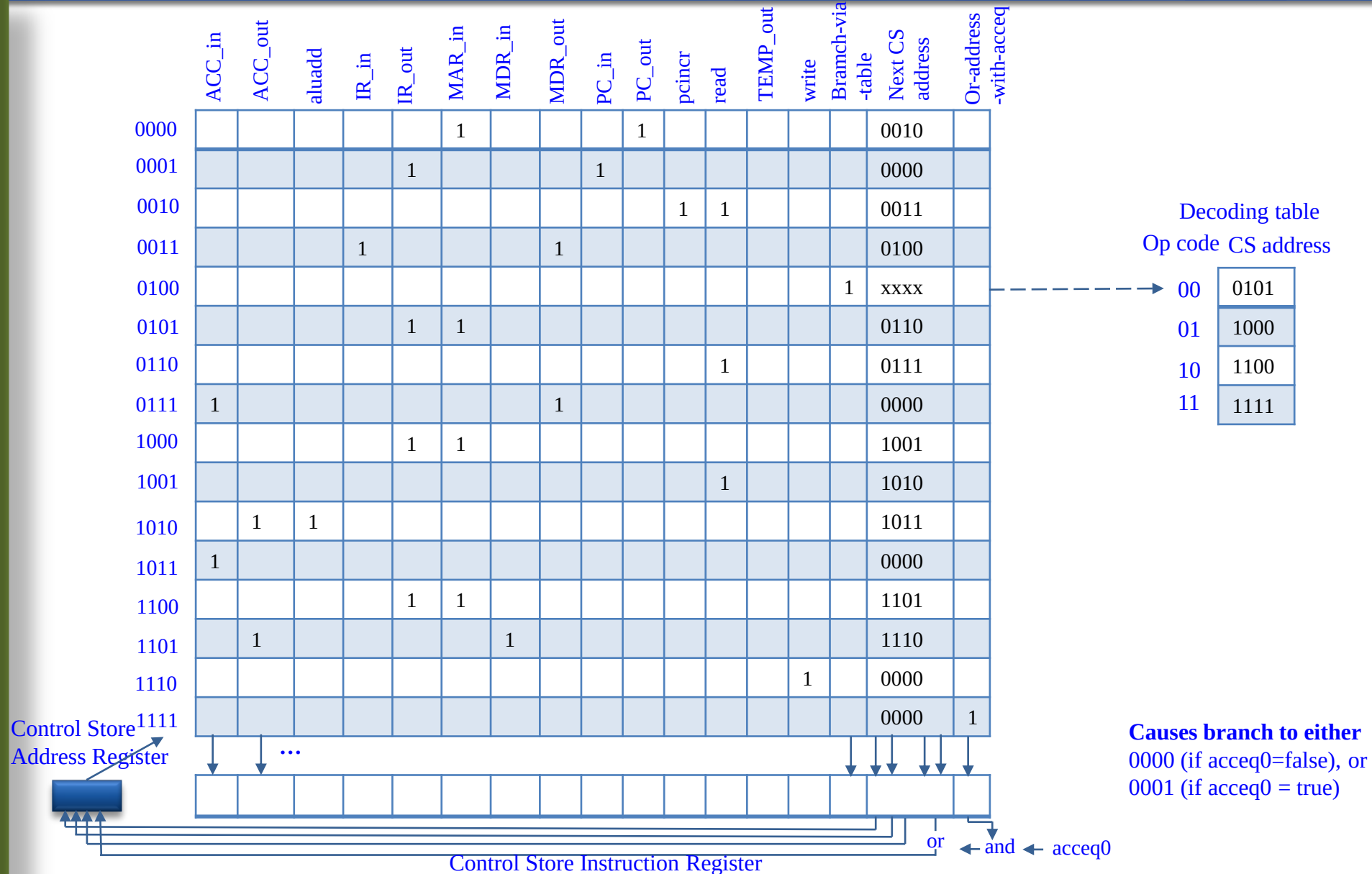
$Grb, R_{out}, C = B$

$n \neq 0 \rightarrow (C_{out}, SHR, C_{in}, Decr, Goto7)$

$C_{out}, Gra, R_{in}, End$

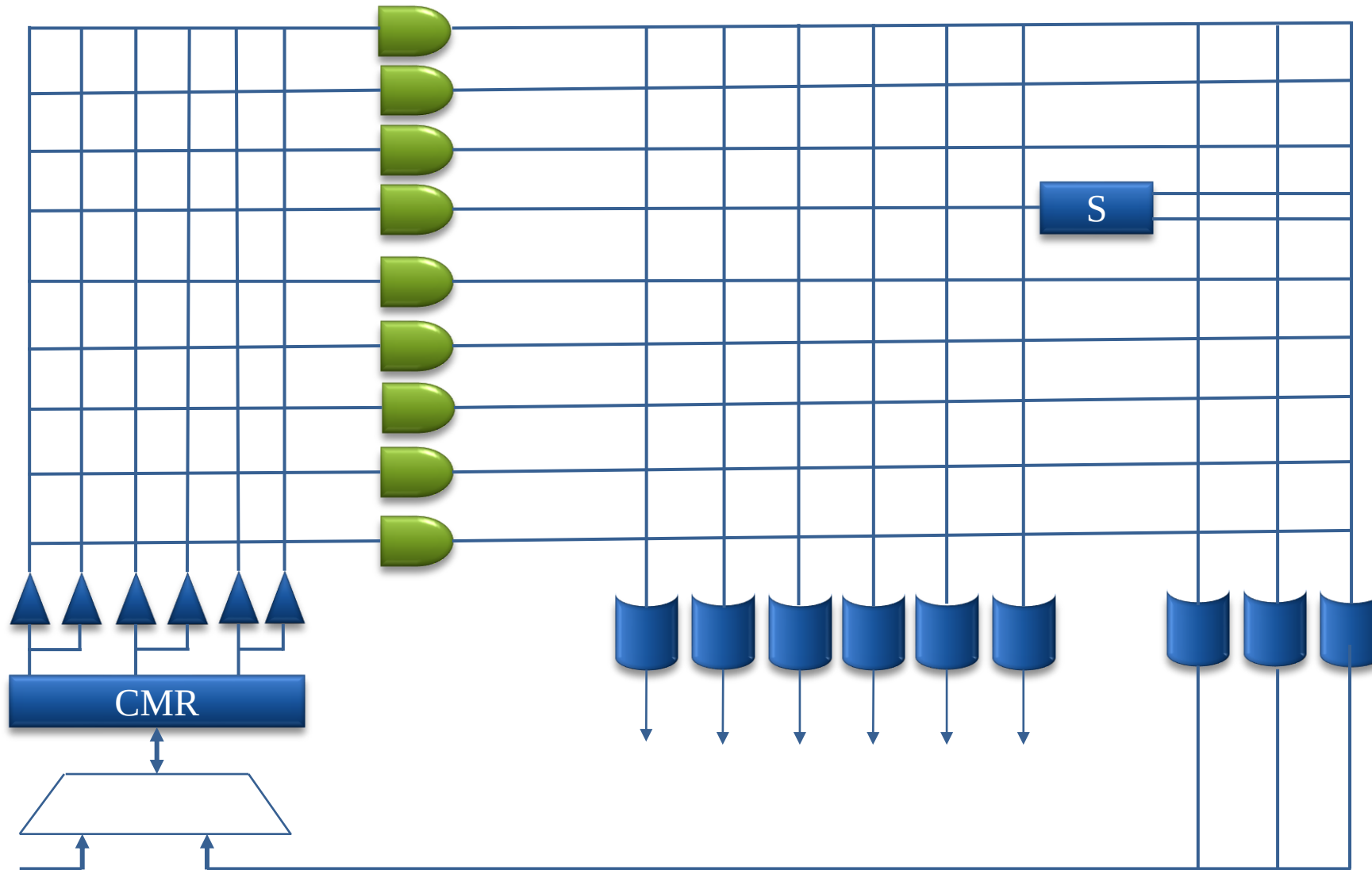
Control Store (16 x20 bits)

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Control store / memory

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- ❑ Microporgrammed control unit pros and cons

Pros

- It simplifies the design of the control unit
- it is both cheaper and less error prone to implement
- decoders and sequencing logic unit of a microprogrammed control unit are very simple pieces of logic
- is the dominant technique for implementing control units in pure CISC architectures, due to its ease of implementation

Cons

- It will be somewhat slower than a hardwired unit of comparable technology

Hardwired Control Unit Vs. Micro-Programmed control unit

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- ❑ Hardwired Control Unit Vs. Micro-Programmed control unit

Hardwired Control Unit

- Hardwired control unit generates the control signals needed for the processor using logic circuits
- Hardwired control unit is faster when compared to microprogrammed control unit as the required control signals are generated with the help of hardwares
- Difficult to modify as the control signals that need to be generated are hard wired
- More costlier as everything has to be realized in terms of logic gates
- It cannot handle complex instructions as the circuit design for it becomes complex
- Only limited number of instructions are used due to the hardware implementation
- Used in computer that makes use of Reduced Instruction Set Computers(RISC)

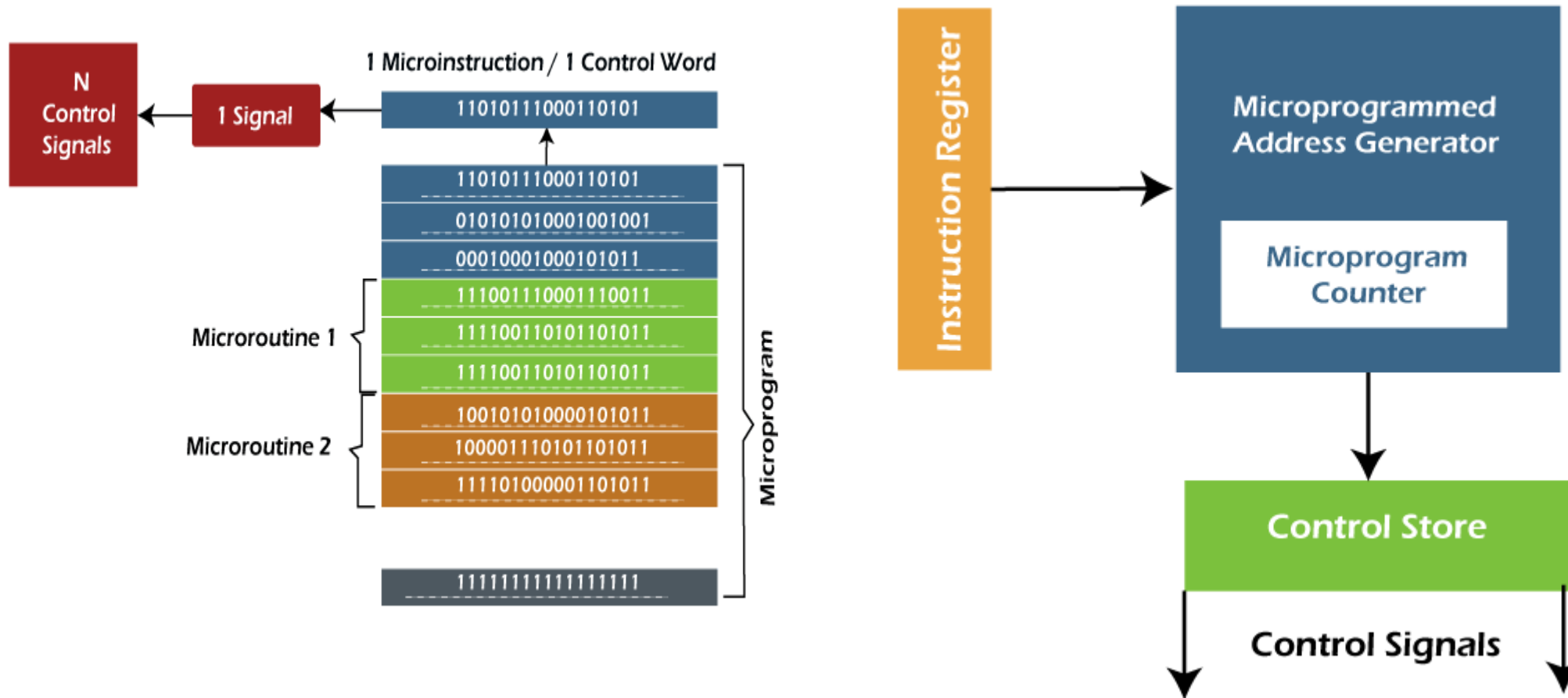
Microprogrammed Control Unit

- Microprogrammed control unit generates the control signals with the help of micro instructions stored in control memory
- This is slower than the other as micro instructions are used for generating signals here
- Easy to modify as the modification need to be done only at the instruction level
- Less costlier than hardwired control as only micro instructions are used for generating control signals
- It can handle complex instructions
- Control signals for many instructions can be generated
- Used in computer that makes use of Complex Instruction Set Computers(CISC)

Summary

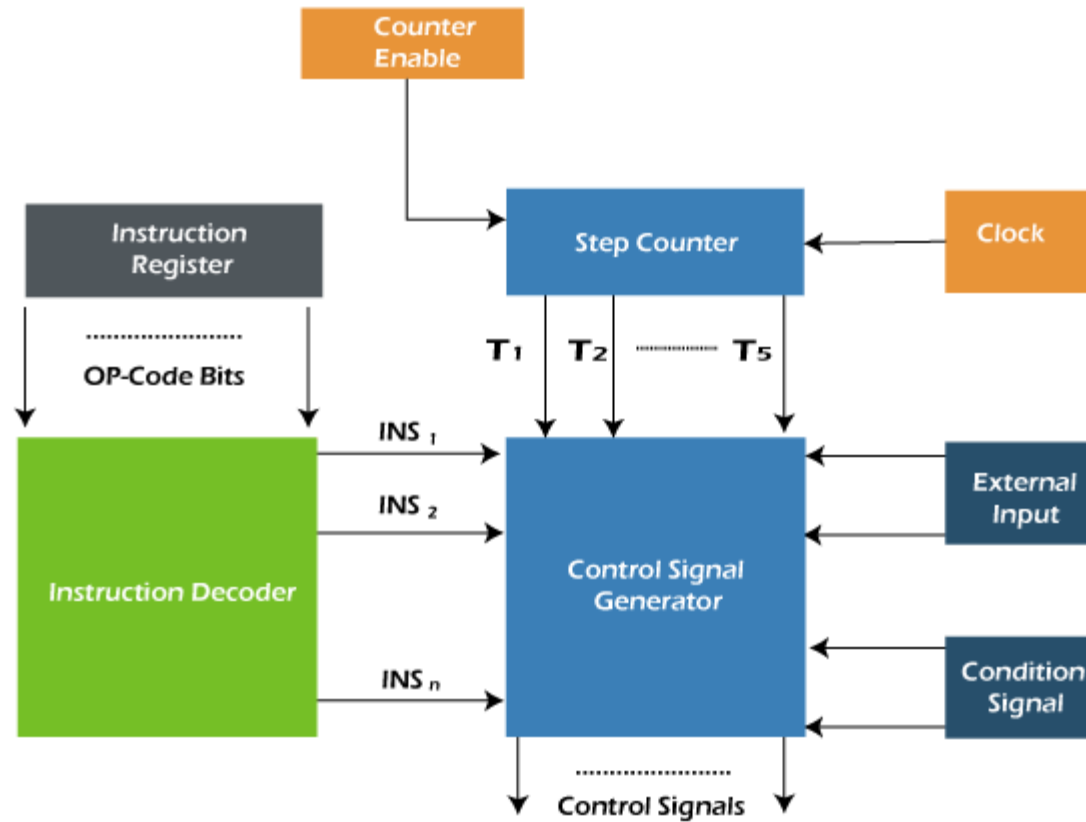
1. Micro Programmed Control Unit
2. Wilke's model
3. Control store design

Microprogrammed Control Unit



- **Micro programmed Control:-** in which control signals are generated by a program similar to machine language programs
- **Control word:** is a word in which each individual bit represents various control signals. Each of the control steps in the control sequence of a instruction defines a unique combination of 1s and 0's of CW
A sequence of CW corresponds to control sequence of machine instruction constitutes the **micro routine of that instruction**. And the individual control word in the micro routine is **micro instructions**
- Micro routines are stored in a special memory called **control store**

Hardwired Control Unit



Hardwired Control Unit

Text Book

- William Stallings “Computer Organization and architecture” Prentice Hall, 7th edition, 2006.

Questions ?



Thank You